Department of Education

SHS-TVL
Computer Programming
JAVA NC III

Module 1

Introduction to Java Programming Language

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What I Need To Know

This module was designed and written with you in mind. It is here to help you **understand the Java Programming Language**. The scope of this module permits it to explore Java Technology – from its short history, its platform editions, its features, the mechanism of a Java Virtual Machine and the change it brought in our life. The Java Programming Language provides an overview of what this technology can do for you.

After going through this module, you are expected to:

1. Describe brief History of Java
2. Define Java Technology and Java programming language;
3. Differentiate Java Platform;
4. Describe Java Virtual Machine (JVM) and Explain the content of Java Development Kit (JDK).

What I know

Directions: Choose the letter of the best answer. Use a separate sheet of paper for your answers.

1. It translates the bytecodes into native language.
   A. JDK B. JVM C. J2SE D. API

2. Java programming language was originally developed by.
   A. Moon Microsystems C. Star Microsystems
   B. Sky Microsystems D. Sun Microsystems

3. Who is the the developer of Java program.
   A. James Gosling B. John Gosling C. Josh Gosling D. James Gossip

4.This is use to translate code into machine language.
   A. compiler C. editor
   B. computer D. translator

5. Which set of Java tools for developing Java programs.
   A. API C. JDK
   B. CDK D. JVM

6. The Java Development Kit consist of the following tools EXCEPT.
A. Hardware B. Java API C. JVM D. Screw

7. It is a simple programming language that is developed by James Gosling and his team.
A. C# B. Java C. C Language D. Cobol

8.It is Java Platform targeting applications running on workstations
A. J2SE B. J2EE C. J2ME D. J2SO

8. It is the prewritten code, organized into packages of similar topics.
A. API B. JVM C. JDK D. IDE

10. Java Development Kit consist of the following EXCEPT.
A. Java API B. Java Compiler C. JVM D. Class

LT 1 Introduction to Java Programming Language

Computer programming language, any of the various languages for a digital computer to communicate a series of comprehensive instructions. Such instructions can be executed directly when they are in the computer-specific numerical form known as the machine language, after a simple process of substitution when expressed in a corresponding assembly language, or after translation from some "higher level" language. While computer languages are numerous, relatively few are commonly used.

A programming language is a vocabulary and set of grammatical rules for instruction to perform specific tasks on a computer or computing device. The term programming language generally refers to high-level languages such as BASIC, C, C++, COBOL, Java, FORTRAN, Ada, Pascal and others.

What's In

A. Guess What?

Directions: Arrange the letters to form a word.

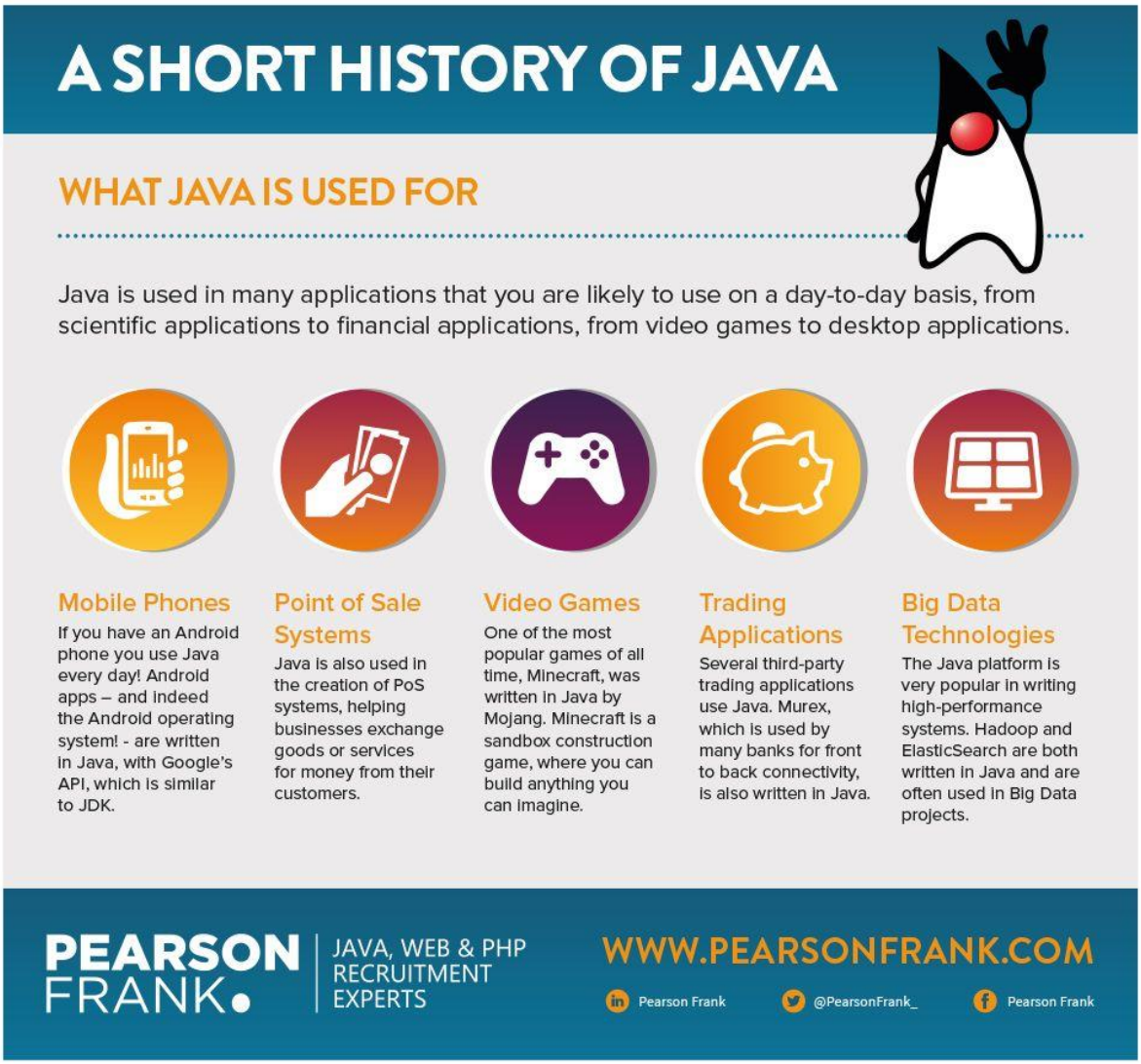
1. **AVJA MROFTALP** _____
2. **OCMIEPLR** _____
3. **AVJA** _____
4. **ODCE** _____
5. **AJAVA ENHCIAM TUALVIR** _____

What's New

Directions: Complete the table below by writing the arranged words in ***What's In*** *in the first column, then give the definition in the second column. This will complete the* ***Five Phases of Java Program****.*

Phases of a Java Program	**Definition**

What Is It



JAVA was developed by Sun Microsystems Inc in 1991, later acquired by Oracle Corporation. It was developed by James Gosling and Patrick Naughton. **It is a simple programming language**. Writing, compiling and debugging a program is easy in java. It helps to create modular programs and reusable code.

Java Technology

Java technology is both a programming language and a platform.

The Java Programming Language

The Java programming language is a high-level language that can be characterized by all of the following buzzwords:

- Simple
- Object oriented
- Distributed
- Multithreaded
- Dynamic
- Architecture neutral
- Portable
- High performance
- Robust
- Secure

Java Platform Editions

A Java Platform is the set of APIs, class libraries, and other programs used in developing Java programs for specific applications.

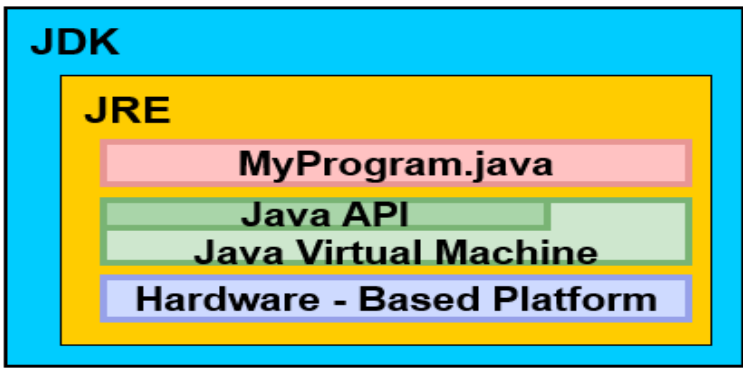
There are 3 Java Platform Editions

1. Java 2 Platform, Standard Edition (J2SE)
    - Core Java Platform targeting applications running on workstations
2. Java 2 Platform, Enterprise Edition (J2EE)
    - Component-based approach to developing distributed, multi-tier enterprise applications
3. Java 2 Platform, Micro Edition (J2ME)
    - Targeted at small, stand-alone or connectable consumer and embedded devices

Java Development Kit (JDK)

Java Development Kit (JDK)

- Is a set of Java tools for developing Java programs
- Consists of Java API, Java Compiler, and JVM



Java Runtime Environment (JRE)

- also known as Java Runtime, is **part** of the Java Development Kit (JDK), ***a set of programming tools for developing Java applications***.

Java Application Programming Interface (API)

- Is prewritten code, organized into packages of similar topics

Java Virtual Machine (JVM)

- Is an execution engine that runs compiled Java byte code

How does JVM work?

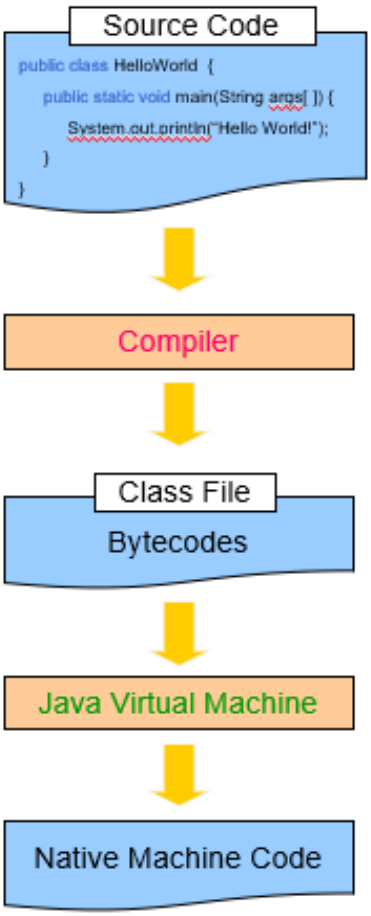
A Java program is written

The program is compiled

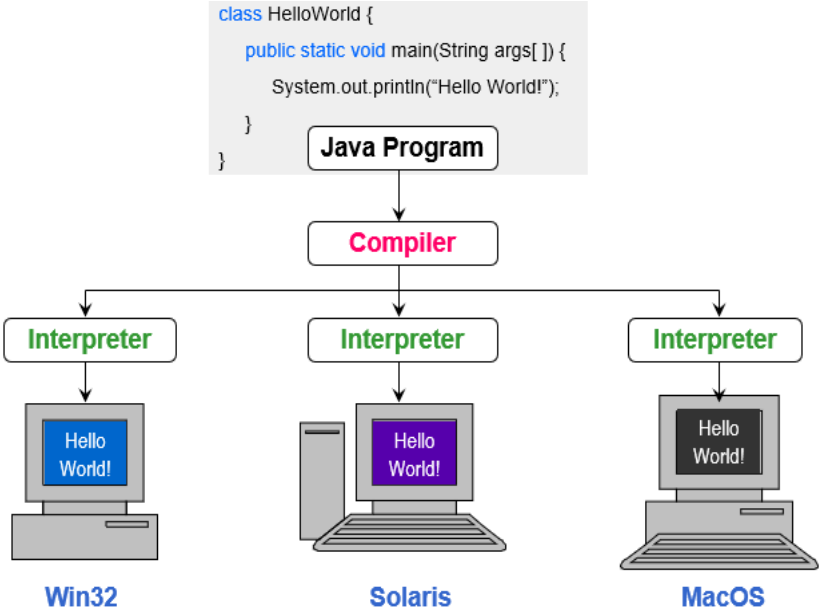
A *class file* is produced containing *bytecodes*

The bytecodes are interpreted by the JVM

The JVM translates bytecodes into native machine code



Running on Different Platforms



Try On

Directions: Base on your understanding explain the Java Development Kit.

What I Have Learned

Directions: Express what you have learned in this lesson by completing the sentences below.

1. Java is a _____
_____
_____.

2. Java Development Kit consist of _____
_____
_____.

3. Java Program runs on different platform such as
_____
_____.

What I Can Do

Direction: Write down the procedure on how to cook rice. Follow the procedure try to do it at home. Explain how are you relate to Java Virtual Machine. **(Application of Java Virtual Machine)**

Directions: Choose the letter of the best answer. Use a separate sheet of paper for your answers.

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6. It is the prewritten code, organized into packages of similar topics.
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7. Java programming language is characterized by the following EXCEPT.
   A. Simple B. Object Oriented C. Not Portable D. Distributed

8. It translates the bytecodes into native language.
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Answer Key

1. A
2. C
3. D
4. B
5. A
6. A
7. C
8. B
9. D
10. A

References

1.https://docs.oracle.com/javase/tutorial/getStarted/intro/definition.html

2. https://beginnersbook.com/2013/05/java-introduction/